

STATISTICS 5 4.1 Probability Distributions

1. The basics

- 1) X = **name** of random variable e.g.: 'days of the week'
 upper case
- 2) Random variables have **no fixed value** - it's all up to chance.
- 3) x = a **certain value** that X can take e.g.: 'Mon, Tues, Wed, Thurs,'
 lower case
- 4) **Discrete random variables** only have a **certain number of possible values**.
 ('drv' → usually whole numbers, but doesn't have to be)

$$\sum P(X = x) = 1$$

total probability = 1

EXAMPLE

The discrete random variable X has the probability distribution shown below.

x	0	1	2	3	4
$P(X = x)$	0.1	0.2	0.3	0.2	a

probability distribution table

Find: (i) the value of a , (ii) $P(2 \leq X < 4)$, (iii) the mode

- (i) Use the formula $\sum_{\text{all } x} P(X = x) = 1$ again.

From the table: $0.1 + 0.2 + 0.3 + 0.2 + a = 1$

$$0.8 + a = 1$$

$$a = 0.2$$

Careful with the inequality signs — you need to include $x = 2$ but not $x = 4$.

- (ii) This is asking for the probability that ' X is greater than or equal to 2, but less than 4'.
Easy — just add up the probabilities.

$$P(2 \leq X < 4) = P(X = 2) + P(X = 3) = 0.3 + 0.2 = 0.5$$

- (iii) The mode is the value of x with the biggest probability — so **mode = 2**.

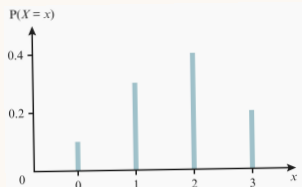
2) Expectation : some drvs have higher or lower probabilities
 ↳ affected the expected outcome or mean of an experiment

Ex.) A biased spinner

possible scores

x	0	1	2	3	
$P(X=x)$	0.1	0.3	0.4	0.2	$\sum P$
xP	0	0.3	0.8	0.6	1.7

$\sum xP$



$$\text{Mean} = E(X) = \frac{\sum xP}{1}$$

$$= \frac{1.7}{1} = 1.7$$

over 1600 trials?

x	0	1	2	3	$\sum$
Expected frequency (f)	$0.1 \times 1600 = 160$	$0.3 \times 1600 = 480$	$0.4 \times 1600 = 640$	$0.2 \times 1600 = 320$	1600
fx	0	480	1280	960	2720

$\sum f$

$\sum fx$

$$\begin{aligned} E(X) &= \frac{\sum fx}{\sum f} \\ &= \frac{2720}{1600} \\ &= 1.7 \end{aligned}$$

3) Variance = the spread of values around the mean $E(X)$

x	0	1	2	3	
$P(X=x)$	0.1	0.3	0.4	0.2	1
xP	0	0.3	0.8	0.6	1.7
x^2P	0	0.3	1.6	1.8	3.7

$xP \times x$

$\sum x^2P$

$$\text{Var}(X) = \sum x^2P - \{E(X)\}^2$$

$$\begin{aligned} \text{Var}(X) &= 3.7 - 1.7^2 \\ &= 4.7 \end{aligned}$$

Binomial + Geometric Distributions

in a **fixed** no. of trials,
how many **successes**?

in an **infinite** no. of trials,
what is the probability of the first **success**?

1. The binomial distribution

- outcomes of a trial $\left\{ \begin{array}{l} \text{success; probability } p \\ \text{failure; probability } q \end{array} \right\} \quad p + q = 1$

Ex.: Samantha sells light bulbs in packs of 3.

- P(good bulb): $p = 0.9$
- P(faulty bulb): $q = 0.1$
- No. of faulty bulbs in a pack of 3: X ← discrete random variable
which could be 0, 1, 2, 3: r

r	0	1	2	3
$P(X=r)$	q^3	$3pq^2$	$3p^2q$	p^3

$${}^3C_1 \cdot p^1 \cdot q^{3-1}$$

binomial distribution of X

$$X \sim B(n, p)$$

of repeated,
independent trials
(finite) probability
of success

$$P_r = \binom{n}{r} p^r (1-p)^{n-r}$$

probability of r successes

for the

above example:

$$X \sim B(3, 0.9)$$

$$\begin{aligned} P_1 &= \binom{3}{1} \times 0.9^1 \times (1-0.9)^{(3-1)} \\ &= 3 \times 0.9^1 \times 0.1^2 \\ &= 0.027 \end{aligned}$$

* Expectation / mean of $X \sim B(n, p)$

$$E(X) = \mu = np$$

e.g.: $X \sim B(3, 0.1)$

$$\begin{aligned} \mu &= n \cdot p \\ &= 3 \times 0.1 = 0.3 \end{aligned}$$

* Variance:

$$\text{Var}(X) = \sigma^2 = n \cdot p \cdot q \text{ or } n \cdot p \cdot (1-p)$$

σ = standard deviation

$$\begin{aligned} \sigma^2 &= 3 \times 0.1 \times 0.9 \\ &= 0.27 \end{aligned}$$

2. The geometric distribution

Ex.: What are the chances of rolling a 6 on a fair die?

- P(on the 1st roll): p ← success
- P(2nd): $(1-p)p$ ← 1 failure followed by a success $q \cdot p$
- P(3rd): $(1-p)^2 p$ ← 2 failures followed by a success $q^2 \cdot p$

The 1st success occurs on the r^{th} trial:

r	1	2	3	4	...	n	...
$P(X=r)$	p	$p(1-p)$	$p(1-p)^2$	$p(1-p)^3$	...	$p(1-p)^{n-1}$	...

- a series of repeated, independent trials (can be infinite)
- only 2 outcomes (e.g.: heads/tails)
- probability of success p is constant

geometric distribution of X

$$X \sim \text{Geo}(p)$$

probability of success

$$P_r = p(1-p)^{r-1} \text{ for } r = 1, 2, 3, \dots$$

r^{th} trial

$$P(X \leq r) = 1 - q^r$$

$$P(\text{success on one of } 1^{\text{st}} r \text{ trials}) = 1 - P(\text{failure on the first } r \text{ trials})$$

$$P(X > r) = q^r$$

$$P(1^{\text{st}} \text{ success after } r^{\text{th}} \text{ trial}) = P(\text{failure on the first } r \text{ trials})$$

* mode is 1 for all geometric distributions

	$P(X=1)$	$P(X=2)$	$P(X=3)$	$P(X=4)$	$P(X=5)$
Geo(0.2)	0.2	0.16	0.128	0.1024	0.08192
Geo(0.7)	0.7	0.21	0.063	0.0189	0.00567

↳ the highest chance of 1st success is on the 1st trial.

* Expectation for $X \sim \text{Geo}(p)$ is

$$E(X) = \mu = \frac{1}{p}$$